

CERTIFICATE OF COMPLIANCE - RESIDENTIAL PERFORMANCE COMPLIANCE METHOD

Project Name: Banh Thien-ADU

Calculation Date/Time: 2023-11-13T06:19:29-08:00

CF1R-PRF-01-E

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Calculation Description: 1060 sf ADU

Input File Name: Banh Thien ADU.rbd22

GENERAL INFORMATION											
01	Project Name		Banh Thien-ADU								
02	Run Title		1060 sf ADU								
03	Project Location		657 Cortez Ave.								
04	City		Vista CA		05	Standards Version		2022			
06	Zip code		92084		07	Software Version		CBECC-Res 2022.2.0			
08	Climate Zone		7		09	Front Orientation (deg/ Cardinal)		0			
10	Building Type		Single family		11	Number of Dwelling Units		1			
12	Project Scope		Newly Constructed		13	Number of Bedrooms		2			
14	Addition Cond. Floor Area (ft ²)		0		15	Number of Stories		1			
16	Existing Cond. Floor Area (ft ²)		n/a		17	Fenestration Average U-factor		0.3			
18	Total Cond. Floor Area (ft ²)		1060		19	Glazing Percentage (%)		12.00%			
20	ADU Bedroom Count		n/a								

COMPLIANCE RESULTS	
01	Building Complies with Computer Performance
02	This building incorporates features that require field testing and/or verification by a certified HERS rater under the supervision of a CEC-approved HERS provider.
03	This building incorporates one or more Special Features shown below

Registration Number: 223-P016614280A-000-000-0000000-0000
CA Building Energy Efficiency Standards - 2022 Residential Compliance

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Report Version: 2022.0.000
Schema Version: rev 20220901

HERS Provider: CalCERTS inc.
Report Generated: 2023-11-13 06:20:26

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ENERGY USE INTENSITY				
01	Standard Design (kBtu/ft ² - yr)	Proposed Design (kBtu/ft ² - yr)	Compliance Margin (kBtu/ft ² - yr)	Margin Percentage
Gross EUI ¹	14.02	18.29	-4.27	-30.46
Net EUI ²	4.34	2.46	1.88	43.32

Notes
1. Gross EUI is Energy Use Total (not including PV) / Total Building Area.
2. Net EUI is Energy Use Total (including PV) / Total Building Area.

REQUIRED PV SYSTEMS											
01	02	03	04	05	06	07	08	09	10	11	12
DC System Size (kWdc)	Exception	Module Type	Array Type	Power Electronics	CFI	Azimuth (deg)	Tilt Input	Array Angle (deg)	Tilt: (x in 12)	Inverter Eff. (%)	Annual Solar Access (%)
3.12	NA	Standard (14-17%)	Fixed	none	true	150-270	n/a	n/a	<=7:12	96	98

BATTERY SYSTEMS						
01	02	03	04	05	06	07
Control	Capacity (kWh)	Charging		Discharging		Round Trip Efficiency
		Charging Efficiency	Charging Rate (kW)	Discharging Efficiency	Discharging Rate (kW)	
Basic	8	0.95	n/a	0.95	n/a	0.9

REQUIRED SPECIAL FEATURES	
The following are features that must be installed as condition for meeting the modeled energy performance for this computer analysis.	
- PV System: 3.12 kWdc - Battery System: 8 kWh - Compact distribution system basic credit	

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ENERGY DESIGN RATINGS	Energy Design Ratings			Compliance Margins		
	Source Energy (EDR1)	Efficiency ¹ EDR (EDR2efficiency)	Total ² EDR (EDR2total)	Source Energy (EDR1)	Efficiency ¹ EDR (EDR2efficiency)	Total ² EDR (EDR2total)
	Standard Design	32	45.1	30.1		
Proposed Design	24.5	44.9	2.1	7.5	0.2	28
RESULT ³ : PASS						
¹ Efficiency EDR includes improvements like a better building envelope and more efficient equipment ² Total EDR includes efficiency and demand response measures such as photovoltaic (PV) system and batteries ³ Building complies when source energy, efficiency and total compliance margins are greater than or equal to zero and unmet load hour limits are not exceeded						
- Standard Design PV Capacity: 1.91 kWdc - Proposed PV kWh output exceeds proposed electricity use by 60% which may violate NEM rules. Contact local utility. - Proposed PV system downsized to 3.12 kWdc (a factor of 0.624) due to cap of 1.6 x proposed design electricity use						

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HERS FEATURE SUMMARY	
The following is a summary of the features that must be field-verified by a certified HERS Rater as a condition for meeting the modeled energy performance for this computer analysis. Additional detail is provided in the building tables below. Registered CF2Rs and CF3Rs are required to be completed in the HERS Registry	
- Quality insulation installation (QII) - Indoor air quality ventilation - Kitchen range hood - Minimum Airflow - Verified SEER/SEER2 - Verified Refrigerant Charge - Fan Efficacy Watts/CFM - Duct leakage testing	

BUILDING - FEATURES INFORMATION						
01	02	03	04	05	06	07
Project Name	Conditioned Floor Area (ft ²)	Number of Dwelling Units	Number of Bedrooms	Number of Zones	Number of Ventilation Cooling Systems	Number of Water Heating Systems
Banh Thien-ADU	1060	1	2	1	0	1

ZONE INFORMATION						
01	02	03	04	05	06	07
Zone Name	Zone Type	HVAC System Name	Zone Floor Area (ft ²)	Avg. Ceiling Height	Water Heating System 1	Status
Conditioned	Conditioned	HVAC System 1	1060	9	DHW System	New

OPAQUE SURFACES							
01	02	03	04	05	06	07	08
Name	Zone	Construction	Azimuth	Orientation	Gross Area (ft ²)	Window and Door Area (ft2)	Tilt (deg)
Front	Conditioned	R15 Stucco Wall	0	Front	234	39	90
Left	Conditioned	R15 Stucco Wall	90	Left	450	48	90
Back	Conditioned	R15 Stucco Wall	180	Back	234	32	90
Right	Conditioned	R15 Stucco Wall	270	Right	450	28	90
Ceiling (below attic) 1	Conditioned	R30 Ceiling below attic	n/a	n/a	1060	n/a	n/a

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ENERGY USE SUMMARY						
Energy Use	Standard Design Source Energy (EDR1) (kBtu/ft ² -yr)	Standard Design TDV Energy (EDR2) (kTDV/ft ² -yr)	Proposed Design Source Energy (EDR1) (kBtu/ft ² -yr)	Proposed Design TDV Energy (EDR2) (kTDV/ft ² -yr)	Compliance Margin (EDR1)	Compliance Margin (EDR2)
Space Heating	0.58	2.6	0.44	2	0.14	0.6
Space Cooling	0.37	8.6	0.21	5.31	0.16	3.29
IAQ Ventilation	0.39	4.19	0.39	4.19	0	0
Water Heating	1.79	19.94	5.79	23.66	-4	-3.72
Self Utilization/Flexibility Credit				0		0
Efficiency Compliance Total	3.13	35.33	6.83	35.16	-3.7	0.17
Photovoltaics	-2.02	-50.9	-3.3	-84.64		
Battery			-4.48	-21.93		
Flexibility						
Indoor Lighting	0.84	8.66	0.84	8.66		
Appl. & Cooking	2.45	20.02	2.45	19.96		
Plug Loads	4.27	45.11	4.27	45.11		
Outdoor Lighting	0.2	1.88	0.2	1.88		
TOTAL COMPLIANCE	8.87	60.1	6.81	4.2		

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ATTIC							
01	02	03	04	05	06	07	08
Name	Construction	Type	Roof Rise (x in 12)	Roof Reflectance	Roof Emittance	Radiant Barrier	Cool Roof
Attic	Tile Roof	Ventilated	5	0.1	0.85	No	No

FENESTRATION / GLAZING													
01	02	03	04	05	06	07	08	09	10	11	12	13	14
Name	Type	Surface	Orientation	Azimuth	Width (ft)	Height (ft)	Mult.	Area (ft ²)	U-factor	U-factor Source	SHGC	SHGC Source	Exterior Shading
10	Window	Front	Front	0	4	4	1	16	0.3	NFRC	0.23	NFRC	Bug Screen
18	Window	Front	Front	0	3	1	1	3	0.3	NFRC	0.23	NFRC	Bug Screen
11	Window	Left	Left	90	4	4	1	16	0.3	NFRC	0.23	NFRC	Bug Screen
12	Window	Left	Left	90	4	4	1	16	0.3	NFRC	0.23	NFRC	Bug Screen
13	Window	Left	Left	90	2	4	1	8	0.3	NFRC	0.23	NFRC	Bug Screen
14	Window	Left	Left	90	2	4	1	8	0.3	NFRC	0.23	NFRC	Bug Screen
15	Window	Back	Back	180	4	4	1	16	0.3	NFRC	0.23	NFRC	Bug Screen
16	Window	Back	Back	180	4	4	1	16	0.3	NFRC	0.23	NFRC	Bug Screen
17	Window	Right	Right	270	2	4	1	8	0.3	NFRC	0.23	NFRC	Bug Screen
19	Window	Right	Right	270	3	4	1	12	0.3	NFRC	0.23	NFRC	Bug Screen
20	Window	Right	Right	270	2	4	1	8	0.3	NFRC	0.23	NFRC	Bug Screen

OPAQUE DOORS			
01	02	03	04
Name	Side of Building	Area (ft ²)	U-factor
115	Front	20	0.2

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